Business Problem Statement

The problem this bootcamp set out to solve

> Bootcamp-generated business case

Date: 2026-11-09 Project: test-claude-plugin-35 Design Pattern: Custom

Problem Description

Customer records are fragmented and duplicated across a CRM, a billing system, and support tickets, each with inconsistent name/address formats and missing fields, making it hard to get a single, trustworthy view of each customer.

See [data architecture and data flow diagrams](data_architecture.md).

Use Case Category

Customer 360

Design Pattern Reference

Not applicable: no design pattern was selected for this scenario.

Data Sources

1. CRM_SYSTEM: Synthetic / ~4,000 records / People / Ongoing / Generated 2. BILLING_SYSTEM: Synthetic / ~4,000 records / People / Ongoing / Generated 3. SUPPORT_TICKETS: Synthetic / ~2,000 records / People / Ongoing / Generated

Entity Types

People

Key Matching Criteria

- Name (High priority): appears as one joined field in CRM, as separate first/last fields in Billing, and as a (sometimes mistyped) name in Support - the primary basis for linking records across sources.
- Address (Medium priority): free-text in CRM, parsed into components in Billing - needed to disambiguate same-name customers.
- Phone (Medium priority): inconsistent formats, sometimes missing - a secondary corroborating attribute.
- Email (Medium priority): present in CRM and Support, sometimes missing - another corroborating attribute, occasionally mistyped in Support.

Success Criteria

- Correctly identify duplicate customer records across CRM, Billing, and Support systems despite inconsistent name/address formats and missing fields.
- Produce a single, unified view of each customer for use in reporting.

- Demonstrate the resolution working end-to-end on the generated, deliberately messy dataset.

Desired Output

Format: Master list, API, Reports Use case: Ongoing Integration: Standalone

Integration Requirements

Downstream systems: None - standalone for now. Integration method: Not applicable. Systems mentioned: None (integration_targets: [] in config/bootcamp_preferences.yaml).

Deployment Target

Platform: Not stated Category: Local Note: development proceeds locally first; infrastructure is a production follow-up covered by the graduation production project and migration checklist.

Timeline

Target completion: Not applicable - bootcamp exercise. Key milestones: Complete each bootcamp module in sequence.

Notes

This is a bootcamp-generated scenario, sized to roughly 10,000 records total across the three sources, deliberately built with cross-source mapping divergence (joined vs. parsed names; free-text vs. parsed addresses) and data-quality variation (missing and off-pattern values) so Data Quality, Mapping, and Transformation has real work to do.